Srivikram MV

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RESEARCH EXPERIENCE

Correct Answers, Inaccurate Source Reports: A Controlled Trace-Transfer Study 2026
Independent Research · Under review at ACL Rolling Review (ARR)

- Developed a controlled chain-of-thought (CoT) evaluation benchmark comprising **280 paired mathematical problems** across seven task families, testing reasoning-report fidelity by transferring fixed, model-generated reasoning traces into counterfactually modified question contexts.
- Evaluated Gemma 4 E2B and GPT-OSS 20B; conflicting questions increased **false trace attributions by 31.7–37.5 percentage points**, even when final answers were correct.

Monitoring Conflict Resolution in Reasoning Language Models 🤗 2026
Independent Research · Imperial College London

- Designed a **120-pair counterfactual benchmark** and **correction-aware taxonomy** of visible repair, silent bypass, transparent recovery and trace adherence; ran **1,080 interventions** across three models at three trace depths.
- Developed **parameter-free candidate-conditioned likelihood scoring** to predict forthcoming answer branches, achieving **ROC AUCs of 1.000, 1.000 and 0.840** across three models on a diagnostic set.
- Compared **2,088 continuations**, hidden-state probes and grounding interfaces; candidate scoring outperformed passive probes, while none of nine interfaces jointly met 80% grounding and policy fidelity.

Auditing Fairness in Large Language Models: Detection and Mitigation Strategies 2025
Independent Research · Newcastle University

- Built a modular bias-auditing pipeline for **four open-weight LLMs (1.3B–14B)** across five intrinsic and extrinsic benchmarks; identified demographic bias and disagreement between evaluation methods.
- Combined **LoRA fine-tuning and inference-time self-debiasing** on Qwen2-1.5B, reducing held-out ambiguous BBQ stereotype selection from **30.80% to 9.82% (68% relative)**.

EDUCATION

Imperial College London 2025–2026
MSc Computing (Artificial Intelligence & Machine Learning)
Selected modules: Deep Learning, Reinforcement Learning, Natural Language Processing, Mathematics for Machine Learning, Machine Learning Systems & Hardware, Deep Graph-Based Learning, SWE-ML, Machine Learning for Imaging.

Newcastle University 2022–2025
*BSc (Hons) Computer Science — **First Class Honours*** | Vice-Chancellor's Excellence Scholar
Selected modules: Computer Vision, Predicative Analysis and AI; Biomedical Data Science & AI.

SELECTED PROJECTS

CUDA Kernel Optimization & ML Systems 2026
Imperial College London

- Developed CUDA C++ inference kernels for batch normalisation and depthwise 2D convolution without PyTorch or cuDNN, optimizing parallel reductions, memory coalescing, and thread-block configurations across varying tensor shapes and convolution parameters.

Transformer Training & Evaluation for Patronising Language Detection 2026
Imperial College London

- Reproduced a SemEval RoBERTa baseline; trained RoBERTa-large with seed sweeps, threshold tuning and ensembling, improving positive-class **F1 from 0.48 to 0.641** and diagnosing lexical shortcuts.

INDUSTRY EXPERIENCE

EAppSys Limited, Oracle Consultancy Partner London, UK
AI & ML Trainee *Jul–Sep 2023*

- Built an OCI/APEX invoice-processing workflow, contributing to document-model annotation and evaluation and integrating LLM-powered natural-language-to-SQL queries.

IBM India Bengaluru, India
Cloud & AI Intern *Jul–Sep 2021*

- Developed IBM Cloud orchestration modules for automated workflows, exploring Watson AI integration via Python and REST APIs and supporting testing and debugging.

TECHNICAL SKILLS

LLM Research: Benchmark design, counterfactual CoT interventions, token likelihood scoring, answer-boundary monitoring, hidden-state probing, model-based annotation, PEFT/LoRA.
Programming & ML: Python, CUDA C++, PyTorch, Hugging Face (Transformers, Datasets, PEFT, Accelerate), vLLM, scikit-learn, TensorFlow/Keras, NumPy, pandas.
Systems: GPU inference optimization, Linux, Git, Docker, Kubernetes, Azure, Oracle Cloud Infrastructure.